

Ques. HOW IS ELECTRICITY GENERATED, AND HOW DO WE USE IT AT HOMES?**Ujjawal R.**

Ans. Straight from the generators, electricity is produced in thousands of volts. This is much more powerful than what we need for everyday use. Even at home, voltage from a wall plug is many times more than most household electronics need or can tolerate. Besides, different components require different voltages and amperages—higher or lower than the current flowing through most of the circuitry. Let us see how we can control this.

At a power station, steam generated from burning coal or heat from a nuclear plant turns a shaft within an electric dynamo. The shaft is attached to a rotor covered with thousands of wires. The rotor turns inside two permanent stator magnets whose fields generate high-voltage electricity in the bushings. This is transmitted as alternating current on the public power grid to homes and businesses.

Before electricity reaches electronic devices, it passes through a step-down transformer, which has a common magnetic core, usually iron, around which two wires are wrapped. Alternating current—electricity that flows first in one direction and then the other—moves through the primary coil—the wire that has more wrappings around the core.

Each time the current switches direction, its magnetic field expands or collapses, which is the same as the field moving, and induces an electric current in the coil. The current passes through the part of the core around which the secondary coil is wrapped.

The field creates a current in the secondary coil, but because the secondary coil has fewer wrappings, current created in the second coil has a smaller voltage, making this a step-down transformer.

Before any electronic appliance like a personal computer can use household current, current passes through another step-down transformer or voltage-reducing circuit.

Resistors are used to decrease the flow of electricity; much like a valve is used to limit the level of water that flows through a pipe. The material and its size changes the amount resistance the component offers. It is measured in ohms.

Variable resistors such as rheostat are also used in situations that call for resistance to be changed manually, on-the-fly. Another example of variable resistor is the potentiometer for controlling volume on a radio receiver.

Q2. PLEASE EXPLAIN THE CONCEPT OF BARCODES AND MATRIX CODES?**Hafsa Rafiqi**

A2. Bar codes. The first six numbers encoded in a bar code identify a company that has paid a fee to acquire the rights to a unique manufacturer identification number. The manufacturer uses the number as the first part of a different 12-digit universal product code (UPC) for each product it makes. The manufacturer's six-digit code stays the same for all its products.

The next five digits signify the item number, or product code. The company's UPC coordinator assigns an item number, not just to every product but to every model or variation of that product.

The last number in the code is a check digit. Its value must match a number obtained by running the other numbers through an algorithm.

In a retail store, when a UPC passes under a laser connected to a register, it does not matter if the right or left side is read first. The bars on the left side are black with white separating these. On the right side, it is reversed—white bars separated by black. This lets the

register know where the code begins.

If the register's computer determines that the check digit does not match the algorithm, the scanner beeps to tell the person at the register to re-scan the UPC.

When a scan is successful, the register sends the number to a computer server. The server checks on the store's current price for that item and sends that price back to the checkout, where it is added to the customer's receipt. For inventory, the server may also update the number of products sold.

Matrix codes. The UPC is good enough if all you need to do is record and read 12 digits. For a verbose code, you need the capacity of a 2D bar code, also called a matrix code or QR code (for quick response). The codes are generally about 645.16mm sq (1-inch-square) in size, but can represent 7000 digits or 4000 characters of text—on average, a little less than 700 words. Using free apps, the codes are read by most smartphones and onscreen by desktops or laptops not equipped with scanners. Not all variety of 2D codes work the same way, though.

Every QR code contains a finder pattern, an arrangement of squares that help the scanner determine the dimensions of the code, the top side and the angle at which the code is being scanned.

A pattern of squares forms an alignment pattern that tells the scanner if the code is distorted.

Along two sides of the code, the scanner reading the code superimposes rows of timing squares, which you do not see. The scanner uses the timing squares to judge how quickly the code is passing through its reading beam.

By calculating the ratio between the light and dark areas in the code, the scanner learns what areas serve to keep every thing lined up and which contain data.

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